

Participant Information Sheet – for parents and guardians

Project title	A visualisation to help school students understand how Large Language Models work
Principal investigator	Prof Judy Robertson– Judy.Robertson@ed.ac.uk
Researcher collecting data	Xinyao Liao – S2810619@ed.ac.uk
Funder	none

This information sheet is for parents and guardians. It explains the research project at the University of Edinburgh in which we would like your child to participate. It gives information about the project in the form of questions you might have and their answers. If you have any further questions, we are happy to discuss them and provide you with more information.

This study has been approved through the Informatics Research Ethics Process (Application No. 282654). Please take time to read the following information carefully. A shorter and age-appropriate version of this information will also be provided to your child if you agree to their participation in the project. You should keep this information sheet for your records.

Who are the researchers?

The researcher collecting data is Xinyao Liao, an MSc Computer Science student at the School of Informatics, University of Edinburgh. This research is being conducted as part of her MSc dissertation and is supervised by Dr Judy Robertson at the University of Edinburgh.

What is the purpose of the study?

The purpose of this study is to evaluate an interactive educational website designed to help young people understand how Large Language Models work. Large Language Models are a type of artificial intelligence technology used in tools such as ChatGPT. The study aims to explore whether the website is understandable, engaging and suitable for young people.

How can my child help?

Your child can help by taking part in a short online evaluation. They will first be asked to read an age-appropriate participant information page and provide their agreement to take part. They will then use the educational website and complete a short online questionnaire about their experience.

The study will be conducted entirely online. There will be no interview, face-to-face meeting, audio recording, video recording or photography.

Does my child have to take part?

No – participation in this study is entirely voluntary and is up to both you and your child. You may decide that your child should not take part, without giving a reason. Your child may also choose not to take part or may stop participating at any time.

Choosing not to participate, or withdrawing from the study, will not affect your or your child's rights in any way. If you or your child wish to withdraw, please contact the researcher. Where possible, we will stop using your child's data in any reports, publications or presentations prepared after the withdrawal request. However, once anonymous responses have been combined with other participants' data, it may no longer be possible to identify and remove an individual response.

What will happen if my child takes part?

If you agree for your child to participate, your child will first use an interactive educational website designed to help young people understand how Large Language Models (LLMs) work. The website contains a series of short interactive activities that can be completed online at the participant's own pace.

After using the website, your child will be invited to complete a short online questionnaire about their experience, including how understandable, engaging and useful they found the website. No interviews or face-to-face meetings will be conducted.

No audio, video or photographs will be collected as part of this study.

Are there any risks associated with taking part?

There are no significant risks associated with participation. Your child may stop participating at any time without giving a reason.

Are there any benefits associated with taking part?

Your child will have the opportunity to learn about Large Language Models (LLMs) through an interactive educational website and contribute to the development of educational resources that may benefit future learners.

What will happen to the results of this study?

The results of this study may be summarised in published articles, reports, presentations and the researcher's MSc dissertation. All questionnaire responses will be anonymised before analysis, and no information that could reasonably identify your child will be included in any publication or presentation.

With your consent, anonymised data may also be used for future research. Your child's data may be retained securely for up to three years in accordance with the University of Edinburgh's data retention policy.

How will personal information be protected?

Your child's data will be processed in accordance with UK Data Protection Law. All information collected about your child will be kept strictly confidential. Your child will be assigned a unique participant number, and their name will not be used during data analysis.

Your child's data will only be accessed by the research team, consisting of **Xinyao Liao**, the MSc researcher conducting this study, and **Prof Judy Robertson**, the project supervisor.

All electronic data will be stored securely on password-protected encrypted devices and on the University of Edinburgh's secure storage services. As this study is conducted entirely online, no paper records will be collected. Consent information will be stored separately from questionnaire responses wherever possible to minimise the risk of identification.

What are my and my child's data protection rights?

The University of Edinburgh is the Data Controller for the information you and your child provide. You have the right to access information held about your child. Your right of access can be exercised in accordance with UK Data Protection Law. You also have other rights, including the rights to correction, erasure and objection. For more details, including the right to lodge a complaint with the Information Commissioner's Office, please visit www.ico.org.uk. Questions, comments and requests about your personal data can also be sent to the University Data Protection Officer at dpo@ed.ac.uk.

Who can I contact?

If you have any further questions about the study, please contact the researcher collecting data, **Xinyao Liao**, at S2810619@ed.ac.uk

If you wish to make a complaint about the study, please contact **inf-ethics@inf.ed.ac.uk**. When contacting us, please provide the study title and details of the nature of your complaint.

Alternative formats

To request this document in an alternative format, such as large print or on coloured paper, please contact the researcher collecting data, **Xinyao Liao**, at S2810619@ed.ac.uk

General information

For general information about how the University of Edinburgh uses your data, please visit:

<https://research-office.ed.ac.uk/about/privacy-notice>

Once again, participation in this study is entirely voluntary. Neither you nor your child is under any obligation to take part. Even if you agree now, you may withdraw your child from the study at any time and for any reason by contacting the researcher. Your child may also choose to stop participating at any time without giving a reason.

Thank you for taking the time to read this.

If you decide to allow your child to take part, please complete the online consent form before your child begins the study. Once you have provided your consent electronically, your child may proceed to use the educational website and complete the online questionnaire.